

Experiment No. 5a

Round Robin Scheduling

Structure Definition:

- struct Process: Defines a structure to represent a process with its attributes:
- pid: Process ID.
- arrival: Arrival time of the process.
- burst: Total burst time of the process.
- remaining: Remaining burst time of the process.
- completion: Completion time of the process.
- turnaround: Turnaround time of the process.
- waiting: Waiting time of the process.

Function swap:

- A helper function to swap two process structures. Used for sorting.

Function sortByArrival:

- Sorts the processes based on their arrival times using a bubble sort algorithm.

Function roundRobin:

- Implements the Round Robin scheduling algorithm.
- currentTime: Keeps track of the current time.
- completed: Counts the number of completed processes.
- sequence: An array to store the process execution sequence.
- seqIndex: Index for the sequence array.
- The main loop iterates until all processes are completed.
- It checks each process if it has arrived and still has remaining time.
- If a process is eligible, it's added to the sequence and its remaining time is updated based on the quantum.
- If a process completes in this cycle, its completion time is calculated.
- If no process is executed in a cycle, the current time is incremented.
- Calculates turnaround time and waiting time for each process.
- Displays the process execution sequence, process details, average turnaround time, and average waiting time.

Main Function:

- Reads the number of processes and their arrival and burst times.
- Sorts the processes by arrival time.
- Calls the roundRobin function to perform scheduling.
- Prints the results.

Experiment No. 5b

Priority Scheduling (Non preemptive)

Header Files

- `#include <stdio.h>`: Includes standard input/output library for input and output operations.
- `#include <limits.h>`: Includes header file for constants like `INT_MAX` used for finding minimum value.

Structure Definition

- `struct Process`: Defines a structure to represent a process with its attributes:
- `pid`: Process ID.
- `arrival`: Arrival time of the process.
- `burst`: Burst time of the process.
- `priority`: Priority value of the process (lower value means higher priority).
- `completion`: Completion time of the process.
- `turnaround`: Turnaround time of the process.
- `waiting`: Waiting time of the process.

Function `sortByArrival and Priority`

- Sorts processes based on arrival time and priority.
- If two processes have the same arrival time, they are sorted based on priority.
- Uses a bubble sort algorithm for sorting.

Function `priorityScheduling`

- Implements the priority scheduling algorithm.
- `currentTime`: Keeps track of the current time.
- `completed`: Counts the number of completed processes.
- `isCompleted`: An array to keep track of completed processes.
- `sequence`: An array to store the sequence of process execution.
- `seqIndex`: Index for the sequence array.
- The main loop iterates until all processes are completed.
- Finds the process with the highest priority (lowest priority value) that has arrived.
- If a process is found, update its completion time, turnaround time, and waiting time.
- Mark the process as completed and add it to the sequence.
- If no process is found, increment current Time.
- Calculates and displays average turnaround time and average waiting time.

Main Function

- Reads the number of processes and their arrival, burst time, and priority.
- Sorts processes by arrival time and priority.
- Calls the priority Scheduling function to perform scheduling.
- Prints the results.

Experiment No. 5c

Priority Scheduling – Preemptive

Structure Definition:

- struct Process: Defines a structure to represent a process with its attributes:
- pid: Process ID.
- arrival: Arrival time of the process.
- burst: Total burst time of the process.
- priority: Priority value of the process (lower value means higher priority).
- remaining: Remaining burst time of the process.
- completion: Completion time of the process.
- turnaround: Turnaround time of the process.
- waiting: Waiting time of the process.

Function preemptive Priority Scheduling:

- Implements the preemptive priority scheduling algorithm.
- current Time: Keeps track of the current time.
- completed: Counts the number of completed processes.
- isCompleted: An array to keep track of completed processes.
- sequence: An array to store the sequence of process execution.
- seqIndex: Index for the sequence array.
- The main loop iterates until all processes are completed.
- Finds the process with the highest priority (lowest priority value) that has arrived and has remaining time.
- If a process is found, it executes for one time unit, decrements its remaining time, and adds it to the sequence.
- If a process completes, its completion time, turnaround time, and waiting time are calculated.
- If no process is found, increment current Time.
- Calculates and displays average turnaround time and average waiting time.

Main Function:

- Reads the number of processes and their arrival, burst time, and priority.
- Initializes the remaining burst time for each process.
- Calls the preemptive Priority Scheduling function to perform scheduling.
- Prints the results.